SoccerGMR SI-Bind: Detailed Component Ablation Study (E1 to E9)

Fine-grained ablation across mechanisms: Mechanism A (Binding & Residual), Mechanism B (Slot Innovation), and Mechanism C (Dominance Refresh) on Standard Test (1,036 queries) and Full Test (2,970 queries).

		Standard Test Split (1,036 queries)				Full-Decodable Test Split (2,970 queries)			
Backbone	**Ablation Method / Configuration**	**mAP (%)**	**Orc-Gap (%)**	**Rej-F1@.5 (%)**	**AUROC (%)**	**mAP (%)**	**Orc-Gap (%)**	**Rej-F1@.5 (%)**	**AUROC (%)**
Moment-DETR	1. E1: Binding-only (Interlayer Anchor)	12.32	**18.82**	75.18	**75.96**	10.34	**17.98**	78.49	**82.38**
	2. E2: Lite Binding + Residual Stream	11.85	22.41	72.40	71.55	10.14	21.20	76.11	80.14
	3. E3: E2 + Slot-Private Innovation (B)	**15.60**	24.26	66.22	73.17	**13.30**	23.02	67.12	80.91
	4. E4: E2 + Uniform Random Refresh (C Control)	12.35	24.02	60.18	75.53	12.13	22.83	61.91	81.77
	5. E5: E2 + Dominance-Aware Refresh (C)	7.95	21.56	75.00	68.32	6.47	21.06	**78.77**	77.80
	6. E6: E2 + Innovation (B) + Dominance (C)	15.21	23.66	65.52	74.32	12.58	22.21	68.83	81.86
	7. E7: E6 - Binding (bind_coef=0 Ablation)	15.21	23.66	65.52	74.32	12.58	22.21	68.83	81.86
	8. E8: Strict Serial A -> B (No Refresh C)	12.44	22.03	**75.57**	73.71	10.51	20.95	78.37	81.21
	9. E9: Serial C -> A -> B (Refresh -> Re-bound -> Nov)			--				--	
BM-DETR	2. E2: Lite Binding + Residual Stream	6.36	18.84	**75.00**	68.63	5.68	18.46	78.80	78.79
	3. E3: E2 + Slot-Private Innovation (B)	7.88	21.63	72.74	71.55	7.08	20.95	76.91	79.45
	4. E4: E2 + Uniform Random Refresh (C Control)	6.48	20.60	**75.00**	69.37	5.95	19.98	**78.91**	79.18
	5. E5: E2 + Dominance-Aware Refresh (C)	7.16	**15.26**	**75.00**	69.37	6.93	**15.16**	78.14	78.92
	6. E6: E2 + Innovation (B) + Dominance (C)	7.72	20.91	73.40	70.18	7.03	20.30	77.47	79.27
	7. E7: E6 - Binding (bind_coef=0 Ablation)	7.72	20.91	73.40	70.18	7.03	20.30	77.47	79.27
	8. E8: Strict Serial A -> B (No Refresh C)	**12.26**	21.71	70.99	**73.09**	**10.92**	20.48	76.14	**80.21**
	9. E9: Serial C -> A -> B (Refresh -> Re-bound -> Nov)			--				--	
UVCOM	1. E1: Binding-only (Interlayer Anchor)	15.31	18.42	74.94	77.68	14.60	17.55	73.74	78.58
	2. E2: Lite Binding + Residual Stream	20.31	**14.46**	72.79	77.04	20.02	**14.54**	70.95	76.70
	3. E3: E2 + Slot-Private Innovation (B)	**20.87**	18.11	69.84	77.23	**20.95**	16.99	69.72	76.14
	4. E4: E2 + Uniform Random Refresh (C Control)	17.30	16.39	74.17	77.79	16.34	15.66	**73.78**	**78.94**
	5. E5: E2 + Dominance-Aware Refresh (C)	20.49	17.55	69.48	**78.13**	19.76	16.74	68.12	75.69
	6. E6: E2 + Innovation (B) + Dominance (C)	18.99	16.30	**75.40**	76.91	17.74	15.55	73.11	75.20
	7. E7: E6 - Binding (bind_coef=0 Ablation)	18.99	16.30	**75.40**	76.91	17.74	15.55	73.11	75.20
	8. E8: Strict Serial A -> B (No Refresh C)	11.64	22.85	71.00	72.64	10.77	21.95	71.33	76.12
	9. E9: Serial C -> A -> B (Refresh -> Re-bound -> Nov)			--				--	
CG-DETR	2. E2: Lite Binding + Residual Stream	14.99	23.42	72.77	73.86	12.68	22.41	75.71	**81.67**
	3. E3: E2 + Slot-Private Innovation (B)	16.29	24.97	64.69	75.17	14.54	23.68	65.94	80.57
	4. E4: E2 + Uniform Random Refresh (C Control)	14.47	**20.87**	75.19	74.40	13.00	**19.67**	**78.33**	81.20
	5. E5: E2 + Dominance-Aware Refresh (C)	13.80	21.77	**75.72**	75.42	13.96	21.14	77.43	81.33
	6. E6: E2 + Innovation (B) + Dominance (C)	**17.38**	22.89	70.85	**75.92**	**15.54**	21.61	73.14	80.94
	7. E7: E6 - Binding (bind_coef=0 Ablation)	**17.38**	22.89	70.85	**75.92**	**15.54**	21.61	73.14	80.94
	8. E8: Strict Serial A -> B (No Refresh C)	15.22	23.40	71.42	74.25	13.36	22.11	73.67	81.33
	9. E9: Serial C -> A -> B (Refresh -> Re-bound -> Nov)			--				--	
BAM-DETR	2. E2: Lite Binding + Residual Stream	22.57	**19.78**	**71.79**	78.04	20.75	**18.73**	**72.80**	**79.91**
	3. E3: E2 + Slot-Private Innovation (B)	22.21	22.18	68.50	77.30	20.34	21.06	67.36	77.80
	5. E5: E2 + Dominance-Aware Refresh (C)	23.52	21.13	67.14	78.18	20.72	19.33	67.02	77.81
	6. E6: E2 + Innovation (B) + Dominance (C)	**23.78**	20.40	68.96	78.37	**21.78**	19.07	69.28	78.69
	8. E8: Strict Serial A -> B (No Refresh C)	22.47	22.63	68.11	**78.67**	21.08	21.08	67.47	78.65
	9. E9: Serial C -> A -> B (Refresh -> Re-bound -> Nov)			--				--	
Sim-DETR	2. E2: Lite Binding + Residual Stream	18.99	21.89	75.11	73.55	16.94	21.13	**77.25**	**79.91**
	3. E3: E2 + Slot-Private Innovation (B)	20.95	18.92	**76.07**	**75.34**	19.58	18.64	75.29	78.62
	5. E5: E2 + Dominance-Aware Refresh (C)	19.14	21.15	75.00	72.22	17.31	20.43	75.04	78.05
	6. E6: E2 + Innovation (B) + Dominance (C)	**21.17**	21.26	74.89	75.12	**20.08**	20.83	73.10	77.83
	8. E8: Strict Serial A -> B (No Refresh C)	15.69	**16.83**	75.00	71.80	14.54	**17.04**	74.23	78.70
	9. E9: Serial C -> A -> B (Refresh -> Re-bound -> Nov)			--				--	
EaTR	2. E2: Lite Binding + Residual Stream	16.32	21.30	**74.58**	75.27	**15.43**	**20.24**	**75.47**	79.21
	3. E3: E2 + Slot-Private Innovation (B)	15.11	23.84	61.97	**76.76**	14.73	22.33	64.05	**82.50**
	5. E5: E2 + Dominance-Aware Refresh (C)	**17.56**	24.33	65.09	76.10	14.96	22.85	66.40	82.30
	6. E6: E2 + Innovation (B) + Dominance (C)	17.12	21.80	67.74	76.23	15.35	20.89	68.66	81.03
	8. E8: Strict Serial A -> B (No Refresh C)	6.27	**21.04**	74.33	70.50	5.34	21.12	73.68	78.74
	9. E9: Serial C -> A -> B (Refresh -> Re-bound -> Nov)			--				--	
QD-DETR	2. E2: Lite Binding + Residual Stream	**16.37**	24.11	60.54	**76.84**	13.06	22.36	63.51	**83.60**
	3. E3: E2 + Slot-Private Innovation (B)	12.71	22.15	72.82	74.74	10.70	21.36	74.96	82.04
	5. E5: E2 + Dominance-Aware Refresh (C)	15.21	20.98	72.08	75.61	12.17	**19.67**	74.36	82.21
	6. E6: E2 + Innovation (B) + Dominance (C)	16.30	**20.65**	**73.71**	76.63	**13.20**	19.84	**75.91**	82.47
	8. E8: Strict Serial A -> B (No Refresh C)	14.53	23.75	64.46	75.85	12.93	22.37	66.69	83.00
	9. E9: Serial C -> A -> B (Refresh -> Re-bound -> Nov)			--				--	

Note: E1 = Binding only (identity across layers); E2 = Lite Binding + Residual; E3 = E2 + Slot-Private Innovation (B); E4 = E2 + Uniform Random Refresh (C Control); E5 = E2 + Dominance Refresh (C); E6 = E2+B+C (Champion Triad); E7 = E6 - Binding; E8 = Strict Serial A -> B without C; E9 = Strict Serial C -> A -> B (Refresh -> Re-bound -> Nov).
Bold numbers indicate best performance among ablation variants within the same backbone. Oracle Gap: lower is better.